

Dionex_COMPASS_May_2023_rerun

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Daily Set up

Read in and Format the raw data - change wd & file names

```
# SULFATE DATA:
```

```
## Read in raw data file from Dionex - copied and saved as a txt
```

```
Sdat <- read.table("Raw Data/COMPASS_Synoptic_CB_MonMon_202305_rerun_S04.txt", sep='\t' , header=T, skip=1)  
head(Sdat)
```

```
##      X          X.1          X.2 IC.S04 IC.S04.1 IC.S04.2 IC.S04.3 IC.S04.4  
## 1 1 Lab Blank      Unknown    n.a.    n.a.    n.a.    n.a.    n.a.  
## 2 2 Lab Blank      Unknown    n.a.    n.a.    n.a.    n.a.    n.a.  
## 3 3 Lab Blank      Unknown    n.a.    n.a.    n.a.    n.a.    n.a.  
## 4 4 Lab Blank      Unknown    n.a.    n.a.    n.a.    n.a.    n.a.  
## 5 5 Standard 1 Calibration Standard 4.177 0.4612 6.67 0.4171 3.5  
## 6 6 Standard 2 Calibration Standard 4.177 0.952 6.84 0.861 7.12
```

```
## Only keep the columns that we need
```

```
Sdat <- Sdat[,c(2,5)] # dont need this here  
head(Sdat)
```

```
##      X.1 IC.S04.1  
## 1 Lab Blank    n.a.  
## 2 Lab Blank    n.a.  
## 3 Lab Blank    n.a.  
## 4 Lab Blank    n.a.  
## 5 Standard 1 0.4612  
## 6 Standard 2 0.952
```

```
## Name the columns correctly
```

```
colnames(Sdat) <- c("Sample_ID", "S04_ppm")  
Sdat$Sample_ID <- as.factor(Sdat$Sample_ID)  
Sdat$S04_ppm <- as.numeric(Sdat$S04_ppm)
```

```
## Warning: NAs introduced by coercion
```

```
Sdat <- as.data.frame(Sdat)
head(Sdat)
```

```
##      Sample_ID S04_ppm
## 1 Lab Blank      NA
## 2 Lab Blank      NA
## 3 Lab Blank      NA
## 4 Lab Blank      NA
## 5 Standard 1  0.4612
## 6 Standard 2  0.9520
```

```
#Chloride data
```

```
## Read in raw data file from Dionex - copied and saved as a txt
```

```
Cldat <- read.table("Raw Data/COMPASS_Synoptic_CB_MonMon_202305_rerun_Cl.txt",sep='\t' , header=T, skip=1)
head(Cldat)
```

```
##      X      X.1      X.2 IC.Cl IC.Cl.1 IC.Cl.2 IC.Cl.3 IC.Cl.4
## 1 1 Lab Blank      Unknown n.a.      n.a.      n.a.      n.a.      n.a.
## 2 2 Lab Blank      Unknown 3.08  0.0009  7.39  0.001  0.01
## 3 3 Lab Blank      Unknown 3.083 0.0006  2.65  0.0007  0.01
## 4 4 Lab Blank      Unknown n.a.      n.a.      n.a.      n.a.      n.a.
## 5 5 Standard 1 Calibration Standard 3.083 4.9142 93.17 5.8263 61.12
## 6 6 Standard 2 Calibration Standard 3.083 9.8779 93.05 11.7113 121.72
```

```
## Only keep the columns that we need
```

```
Cldat <- Cldat[,c(2,5)]
head(Cldat)
```

```
##      X.1 IC.Cl.1
## 1 Lab Blank  n.a.
## 2 Lab Blank  0.0009
## 3 Lab Blank  0.0006
## 4 Lab Blank  n.a.
## 5 Standard 1  4.9142
## 6 Standard 2  9.8779
```

```
## Name the columns correctly
```

```
colnames(Cldat) <- c("Sample_ID", "Cl_ppm")
Cldat$Sample_ID <- as.factor(Cldat$Sample_ID)
Cldat$Cl_ppm <- as.numeric(Cldat$Cl_ppm)
```

```
## Warning: NAs introduced by coercion
```

```
Cldat <- as.data.frame(Cldat)
head(Cldat)
```

```
##      Sample_ID Cl_ppm
## 1 Lab Blank      NA
## 2 Lab Blank 0.0009
## 3 Lab Blank 0.0006
## 4 Lab Blank      NA
## 5 Standard 1 4.9142
## 6 Standard 2 9.8779
```

```
## Bring the data back together:
all_dat <- merge(Sdat, Cldat, by="Sample_ID")
head(all_dat)
```

```
##      Sample_ID   S04_ppm   Cl_ppm
## 1           1086.6977 8687.0886
## 2           1086.6977 2764.0886
## 3           1086.6977    0.0006
## 4           1086.6977 2633.3361
## 5           1086.6977      NA
## 6           341.4432 8687.0886
```

```
## Remove empty lines
all_dat <- all_dat[!(is.na(all_dat$Sample_ID) | all_dat$Sample_ID==""), ]
head(all_dat)
```

```
##              Sample_ID S04_ppm Cl_ppm
## 26      1_GCW_202305_UP_LysA_20cm 1.1230 0.7193
## 27     10_GCW_202305_TR_LysB_45cm 3.7603 3.1051
## 28 10_GCW_202305_TR_LysB_45cm_dup 3.8851 3.1684
## 29     11_GCW_202305_TR_LysC_10cm 2.4952 4.8296
## 30 11_GCW_202305_TR_LysC_10cm_spk 92.0921 7.9818
## 31     12_GCW_202305_TR_LysC_20cm 3.4180 6.8751
```

```
all_dat <- droplevels(all_dat[!all_dat$Sample_ID == 'Lab Blank',])
head(all_dat)
```

```
##              Sample_ID S04_ppm Cl_ppm
## 26      1_GCW_202305_UP_LysA_20cm 1.1230 0.7193
## 27     10_GCW_202305_TR_LysB_45cm 3.7603 3.1051
## 28 10_GCW_202305_TR_LysB_45cm_dup 3.8851 3.1684
## 29     11_GCW_202305_TR_LysC_10cm 2.4952 4.8296
## 30 11_GCW_202305_TR_LysC_10cm_spk 92.0921 7.9818
## 31     12_GCW_202305_TR_LysC_20cm 3.4180 6.8751
```

#Note from Steph: Check all sample IDs from MSM Upland in 2023 May samples. Samples were labeled LysA, but are actually LysC from this month and zone only and need documented. If there are samples from MSM UP LysA 10cm listed, then the samples were NOT corrected and that would mean that all of the LysA samples should be LysC and we should document and correct. We know this because we hardly ever get water from LysA 10cm at MSM. This is a one time issue from May 2023 as far as I am aware.

```
## [1] "The string 'MSM_202305_UP_LysA_10cm' was not found in the dataframe."
```

Pull out standards - could do some checks here if we want

```
stds <- all_dat[grep("Standard", all_dat$Sample_ID),]
head(stds)
```

```
##      Sample_ID S04_ppm Cl_ppm
## 311 Standard 1  0.4634 4.9318
## 312 Standard 1  0.4634 5.0684
## 313 Standard 1  0.4634 4.9142
## 314 Standard 1  0.4634 5.0035
## 315 Standard 1  0.4655 4.9318
## 316 Standard 1  0.4655 5.0684
```

```
stds_chk_S <- stds %>%
  group_by(Sample_ID) %>%
  summarise(mean = mean(S04_ppm), sd = sd(S04_ppm))
stds_chk_S$cv <- (stds_chk_S$sd/stds_chk_S$mean)*100
stds_chk_S$flag <- ifelse(stds_chk_S$cv < 2, 'YES', 'NO, rerun')
head(stds_chk_S)
```

```
## # A tibble: 5 x 5
##   Sample_ID   mean     sd     cv flag
##   <fct>       <dbl>   <dbl> <dbl> <chr>
## 1 Standard 1  0.463 0.00161 0.347 YES
## 2 Standard 2  0.958 0.00399 0.417 YES
## 3 Standard 3  1.95  0.00323 0.166 YES
## 4 Standard 4 10.0   0.0168  0.168 YES
## 5 Standard 5 20.0   0.0140  0.0701 YES
```

```
stds_chk_Cl <- stds %>%
  group_by(Sample_ID) %>%
  summarise(mean = mean(Cl_ppm), sd = sd(Cl_ppm))
stds_chk_Cl$cv <- (stds_chk_Cl$sd/stds_chk_Cl$mean)*100
stds_chk_Cl$flag <- ifelse(stds_chk_Cl$cv < 2, 'YES', 'NO, rerun')
head(stds_chk_Cl)
```

```
## # A tibble: 5 x 5
##   Sample_ID   mean     sd     cv flag
##   <fct>       <dbl>   <dbl> <dbl> <chr>
## 1 Standard 1  4.98 0.0633 1.27 YES
## 2 Standard 2 10.0 0.106  1.06 YES
## 3 Standard 3 19.9 0.102  0.513 YES
## 4 Standard 4 101.  0.142  0.141 YES
## 5 Standard 5 200.  0.0832 0.0417 YES
```

Calculate mmol/L concentrations

```
#remove standards from sample dataframe
sampledat <- all_dat[!grepl("Standard", all_dat$Sample_ID),]
head(sampledat)
```

```
##      Sample_ID S04_ppm Cl_ppm
## 26      1_GCW_202305_UP_LysA_20cm 1.1230 0.7193
## 27     10_GCW_202305_TR_LysB_45cm 3.7603 3.1051
## 28 10_GCW_202305_TR_LysB_45cm_dup 3.8851 3.1684
```

```
## 29      11_GCW_202305_TR_LysC_10cm  2.4952 4.8296
## 30 11_GCW_202305_TR_LysC_10cm_spk 92.0921 7.9818
## 31      12_GCW_202305_TR_LysC_20cm  3.4180 6.8751
```

```
# Constants needed for calculations:
```

```
clmw <- 35.45      #molecular weight of Chloride: 35.45
```

```
smw <- 32.06      #molecular weight of sulfur: 32.06
```

```
# Convert ppm to mmol/L
```

```
sampledat$S04_mM <- (sampledat$S04_ppm / smw)
```

```
sampledat$Cl_mM <- (sampledat$Cl_ppm / clmw)
```

```
# Calculate Salinity
```

```
# calculated using the Knudsen equation
```

```
# Salinity = 0.03 + 1.8050 * Chlorinity
```

```
# Ref: A Practical Handbook of Seawater Analysis by Strickland & Parsons (P. 11)
```

```
# =((1.807*Cl_ppm)+0.026)/1000
```

```
sampledat$salinity <- ((1.8070 * sampledat$Cl_ppm) + 0.026) / 1000
```

```
head(sampledat)
```

```
##              Sample_ID S04_ppm Cl_ppm      S04_mM      Cl_mM
## 26      1_GCW_202305_UP_LysA_20cm  1.1230 0.7193 0.03502807 0.02029055
## 27      10_GCW_202305_TR_LysB_45cm  3.7603 3.1051 0.11728946 0.08759097
## 28 10_GCW_202305_TR_LysB_45cm_dup  3.8851 3.1684 0.12118216 0.08937659
## 29      11_GCW_202305_TR_LysC_10cm  2.4952 4.8296 0.07782907 0.13623695
## 30 11_GCW_202305_TR_LysC_10cm_spk 92.0921 7.9818 2.87249220 0.22515656
## 31      12_GCW_202305_TR_LysC_20cm  3.4180 6.8751 0.10661260 0.19393794
##      salinity
## 26 0.001325775
## 27 0.005636916
## 28 0.005751299
## 29 0.008753087
## 30 0.014449113
## 31 0.012449306
```

Pull out dups and check with percent difference

```
#Show me the data that we have from the calculations
```

```
head(sampledat)
```

```
##              Sample_ID S04_ppm Cl_ppm      S04_mM      Cl_mM
## 26      1_GCW_202305_UP_LysA_20cm  1.1230 0.7193 0.03502807 0.02029055
## 27      10_GCW_202305_TR_LysB_45cm  3.7603 3.1051 0.11728946 0.08759097
## 28 10_GCW_202305_TR_LysB_45cm_dup  3.8851 3.1684 0.12118216 0.08937659
## 29      11_GCW_202305_TR_LysC_10cm  2.4952 4.8296 0.07782907 0.13623695
## 30 11_GCW_202305_TR_LysC_10cm_spk 92.0921 7.9818 2.87249220 0.22515656
## 31      12_GCW_202305_TR_LysC_20cm  3.4180 6.8751 0.10661260 0.19393794
```

```
##      salinity
## 26 0.001325775
## 27 0.005636916
## 28 0.005751299
## 29 0.008753087
## 30 0.014449113
## 31 0.012449306
```

```
#pull out any rows that have "dup" in the SampleID column
dups <- sampledats %>%
  filter(str_detect(Sample_ID, "dup")) #have to change this to match data
head(dups)
```

```
##           Sample_ID  S04_ppm  Cl_ppm  S04_mM  Cl_mM
## 1 10_GCW_202305_TR_LysB_45cm_dup  3.8851  3.1684  0.1211822  0.08937659
## 2 20_GCW_202305_WC_LysC_10cm_dup 406.6028 3510.8722 12.6825577 99.03729760
## 3 30_MSM_202305_UP_LysB_45cm_dup 213.5421 1216.8528  6.6607018 34.32588999
## 4 40_MSM_202305_TR_LysC_10cm_dup 338.9721 2798.1289 10.5730536 78.93170381
## 5 50_MSM_202305_WC_LysC_20cm_dup 746.5368 7148.3849 23.2856145 201.64696474
## 6 60_MSM_202305_RHZ_UP_SF6_dup  31.9840  629.7312  0.9976294 17.76392666
##      salinity
## 1  0.005751299
## 2  6.344172065
## 3  2.198879010
## 4  5.056244922
## 5 12.917157514
## 6  1.137950278
```

```
#remove these from sample dataframe in a new dataframe
sampledat2 <- sampledats %>%
  filter(!str_detect(Sample_ID, "dup")) %>%
  filter(!str_detect(Sample_ID, "spk"))
head(sampledat2)
```

```
##           Sample_ID  S04_ppm  Cl_ppm  S04_mM  Cl_mM
## 1  1_GCW_202305_UP_LysA_20cm  1.1230  0.7193  0.03502807  0.02029055
## 2 10_GCW_202305_TR_LysB_45cm  3.7603  3.1051  0.11728946  0.08759097
## 3 11_GCW_202305_TR_LysC_10cm  2.4952  4.8296  0.07782907  0.13623695
## 4 12_GCW_202305_TR_LysC_20cm  3.4180  6.8751  0.10661260  0.19393794
## 5 13_GCW_202305_TR_LysC_45cm 16.4178  6.8918  0.51209607  0.19440903
## 6 14_GCW_202305_WC_LysA_10cm 500.9219 2695.9261 15.62451341 76.04869111
##      salinity
## 1 0.001325775
## 2 0.005636916
## 3 0.008753087
## 4 0.012449306
## 5 0.012479483
## 6 4.871564463
```

```
#remove the dup from these IDs so we will have duplicate sample names
dups$Sample_ID<-gsub("_dup", "", as.character(dups$Sample_ID))
dups <- dups[ , -c(2,3)]
```

```
colnames(dups) <- c('Sample_ID', 'S04_mM_dup', 'Cl_mM_dup', 'salinity_dup')
head(dups)
```

```
##           Sample_ID S04_mM_dup Cl_mM_dup salinity_dup
## 1 10_GCW_202305_TR_LysB_45cm 0.1211822 0.08937659 0.005751299
## 2 20_GCW_202305_WC_LysC_10cm 12.6825577 99.03729760 6.344172065
## 3 30_MSM_202305_UP_LysB_45cm 6.6607018 34.32588999 2.198879010
## 4 40_MSM_202305_TR_LysC_10cm 10.5730536 78.93170381 5.056244922
## 5 50_MSM_202305_WC_LysC_20cm 23.2856145 201.64696474 12.917157514
## 6 60_MSM_202305_RHZ_UP_SF6 0.9976294 17.76392666 1.137950278
```

```
#put it back together with the old data set and look for duplicates
QAdups <- merge(sampled2, dups)
head(QAdups)
```

```
##           Sample_ID S04_ppm Cl_ppm S04_mM Cl_mM
## 1 10_GCW_202305_TR_LysB_45cm 3.7603 3.1051 0.1172895 0.08759097
## 2 20_GCW_202305_WC_LysC_10cm 401.9731 3340.5189 12.5381503 94.23184485
## 3 30_MSM_202305_UP_LysB_45cm 218.3943 1465.2734 6.8120493 41.33352327
## 4 40_MSM_202305_TR_LysC_10cm 320.4288 2628.3192 9.9946600 74.14158533
## 5 50_MSM_202305_WC_LysC_20cm 738.3430 7018.5516 23.0300374 197.98453032
## 6 60_MSM_202305_RHZ_UP_SF6 32.5617 635.5395 1.0156488 17.92777151
##           salinity S04_mM_dup Cl_mM_dup salinity_dup
## 1 0.005636916 0.1211822 0.08937659 0.005751299
## 2 6.036343652 12.6825577 99.03729760 6.344172065
## 3 2.647775034 6.6607018 34.32588999 2.198879010
## 4 4.749398794 10.5730536 78.93170381 5.056244922
## 5 12.682548741 23.2856145 201.64696474 12.917157514
## 6 1.148445876 0.9976294 17.76392666 1.137950278
```

```
QAdups$S04_dups_chk <- ((abs(QAdups$S04_mM-QAdups$S04_mM_dup))/((QAdups$S04_mM+QAdups$S04_mM_dup)/2))*100
QAdups$S04_dups_flag <- ifelse(QAdups$S04_dups_chk <10, 'YES', 'NO, rerun')

QAdups$Cl_dups_chk <- ((abs(QAdups$Cl_mM-QAdups$Cl_mM_dup))/((QAdups$Cl_mM+QAdups$Cl_mM_dup)/2))*100
QAdups$Cl_dups_flag <- ifelse(QAdups$Cl_dups_chk <10, 'YES', 'NO, rerun')

head(QAdups)
```

```
##           Sample_ID S04_ppm Cl_ppm S04_mM Cl_mM
## 1 10_GCW_202305_TR_LysB_45cm 3.7603 3.1051 0.1172895 0.08759097
## 2 20_GCW_202305_WC_LysC_10cm 401.9731 3340.5189 12.5381503 94.23184485
## 3 30_MSM_202305_UP_LysB_45cm 218.3943 1465.2734 6.8120493 41.33352327
## 4 40_MSM_202305_TR_LysC_10cm 320.4288 2628.3192 9.9946600 74.14158533
## 5 50_MSM_202305_WC_LysC_20cm 738.3430 7018.5516 23.0300374 197.98453032
## 6 60_MSM_202305_RHZ_UP_SF6 32.5617 635.5395 1.0156488 17.92777151
##           salinity S04_mM_dup Cl_mM_dup salinity_dup S04_dups_chk S04_dups_flag
## 1 0.005636916 0.1211822 0.08937659 0.005751299 3.264708 YES
## 2 6.036343652 12.6825577 99.03729760 6.344172065 1.145149 YES
## 3 2.647775034 6.6607018 34.32588999 2.198879010 2.246720 YES
## 4 4.749398794 10.5730536 78.93170381 5.056244922 5.624287 YES
## 5 12.682548741 23.2856145 201.64696474 12.917157514 1.103631 YES
## 6 1.148445876 0.9976294 17.76392666 1.137950278 1.790050 YES
```

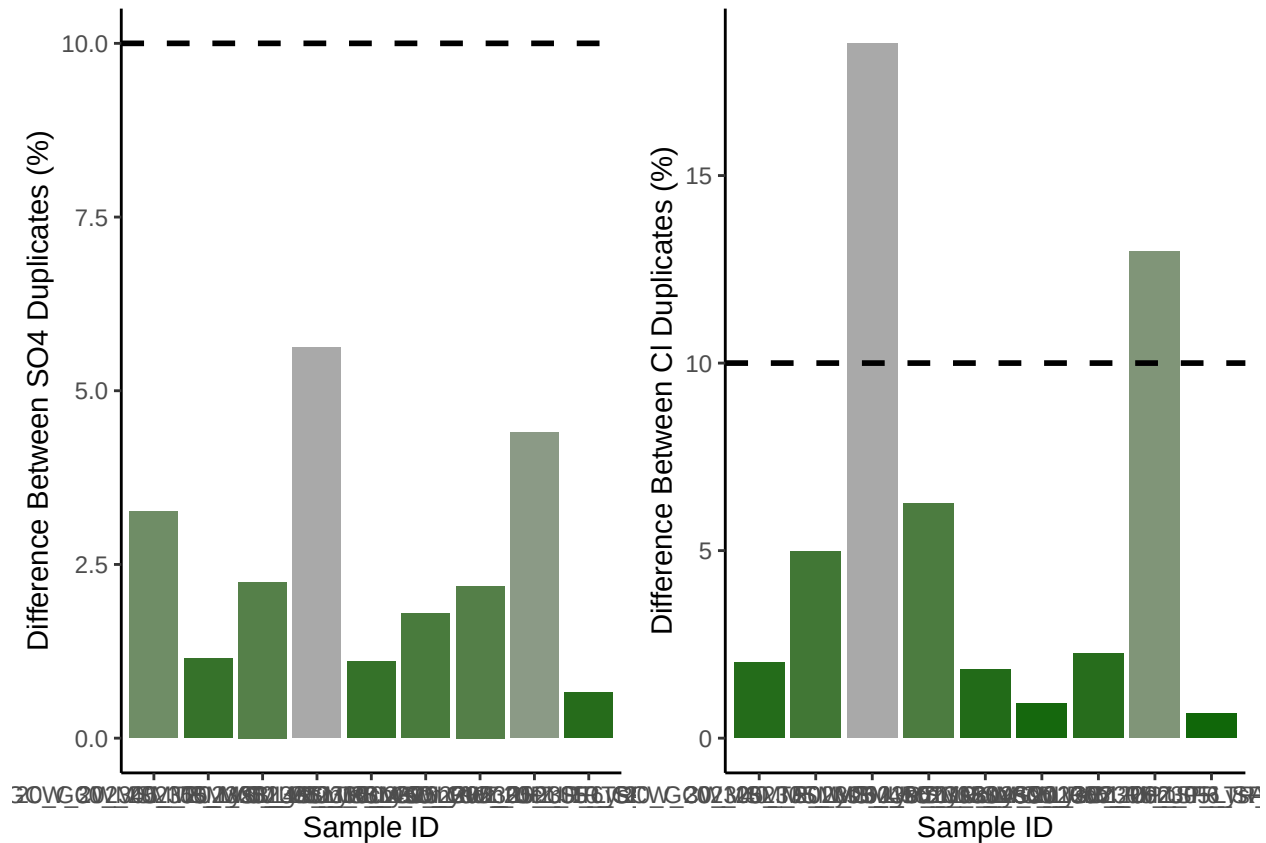
```
##   Cl_dups_chk Cl_dups_flag
## 1   2.0180123          YES
## 2   4.9728091          YES
## 3  18.5241545      NO, rerun
## 4   6.2585948          YES
## 5   1.8329058          YES
## 6   0.9181118          YES
```

```
#plot dups output as a bar graph to easily check - want any over 10% to be red need to work on this
Sdupsbar <- ggplot(data = QAdups, aes(x = Sample_ID, y = S04_dups_chk, fill=S04_dups_chk)) +
  geom_bar(stat = 'identity') +
  scale_fill_gradient2(low='darkgreen', mid='darkgreen', high='darkgrey', space='Lab') +
  theme_classic() + labs(x= "Sample ID", y="Difference Between S04 Duplicates (%)") +
  theme(legend.position="none") + geom_hline(yintercept=10, linetype="dashed",
  color = "black", size=1)
```

```
## Warning: Using 'size' aesthetic for lines was deprecated in ggplot2 3.4.0.
## i Please use 'linewidth' instead.
## This warning is displayed once every 8 hours.
## Call 'lifecycle::last_lifecycle_warnings()' to see where this warning was
## generated.
```

```
Cldupsbar <- ggplot(data = QAdups, aes(x = Sample_ID, y = Cl_dups_chk, fill=Cl_dups_chk)) +
  geom_bar(stat = 'identity') +
  scale_fill_gradient2(low='darkgreen', mid='darkgreen', high='darkgrey', space='Lab') +
  theme_classic() + labs(x= "Sample ID", y="Difference Between Cl Duplicates (%)") +
  theme(legend.position="none") + geom_hline(yintercept=10, linetype="dashed",
  color = "black", size=1)

ggarrange(Sdupsbar, Cldupsbar, ncol=2, nrow=1)
```

```
#check for percent of no, reruns to see if it would warrant reruns
Perc_dups <- QAdups %>%
  group_by(SO4_dups_flag) %>%
  summarise(S_no_rows = length(SO4_dups_flag))
Perc_dups1 <- QAdups %>%
  group_by(Cl_dups_flag) %>%
  summarise(Cl_no_rows = length(Cl_dups_flag))
colnames(Perc_dups) <- c("Flag", "S_no_rows")
colnames(Perc_dups1) <- c("Flag", "Cl_no_rows")
Perc_dups2 <- cbind(Perc_dups, Perc_dups1)

Perc_dups2$Total <- length(QAdups$SO4_dups_flag)
Perc_dups2$S_Percent <- (Perc_dups2$S_no_rows / Perc_dups2$Total)*100
Perc_dups2$Cl_Percent <- (Perc_dups2$Cl_no_rows / Perc_dups2$Total)*100
head(Perc_dups2)
```

```
##   Flag S_no_rows   Flag Cl_no_rows Total S_Percent Cl_Percent
## 1  YES         9 NO, rerun         2     9      100    22.22222
## 2  YES         9   YES           7     9      100    77.77778
```

Pull out dups and check with cv

```

#the cv
# calculate the sd bewteen the two duplicates then divide by the mean and multiply by 100

df2 <- as.data.frame(QAdups$S04_mM)
df2$dups <- QAdups$S04_mM_dup

df2$sds <- apply(df2,1,sd)

QAdups$S04_dups_cv <- (df2$sds)/((QAdups$S04_mM+QAdups$S04_mM_dup)/2) * 100
QAdups$S04_dups_cv_flag <- ifelse(QAdups$S04_dups_cv <11, 'YES', 'NO, rerun')

head(QAdups)

```

```

##           Sample_ID  S04_ppm  Cl_ppm  S04_mM  Cl_mM
## 1 10_GCW_202305_TR_LysB_45cm  3.7603  3.1051  0.1172895  0.08759097
## 2 20_GCW_202305_WC_LysC_10cm 401.9731 3340.5189 12.5381503 94.23184485
## 3 30_MSM_202305_UP_LysB_45cm 218.3943 1465.2734 6.8120493 41.33352327
## 4 40_MSM_202305_TR_LysC_10cm 320.4288 2628.3192 9.9946600 74.14158533
## 5 50_MSM_202305_WC_LysC_20cm 738.3430 7018.5516 23.0300374 197.98453032
## 6 60_MSM_202305_RHZ_UP_SF6 32.5617 635.5395 1.0156488 17.92777151
##           salinity S04_mM_dup  Cl_mM_dup salinity_dup S04_dups_chk S04_dups_flag
## 1 0.005636916 0.1211822 0.08937659 0.005751299 3.264708 YES
## 2 6.036343652 12.6825577 99.03729760 6.344172065 1.145149 YES
## 3 2.647775034 6.6607018 34.32588999 2.198879010 2.246720 YES
## 4 4.749398794 10.5730536 78.93170381 5.056244922 5.624287 YES
## 5 12.682548741 23.2856145 201.64696474 12.917157514 1.103631 YES
## 6 1.148445876 0.9976294 17.76392666 1.137950278 1.790050 YES
##           Cl_dups_chk Cl_dups_flag S04_dups_cv S04_dups_cv_flag
## 1 2.0180123 YES 2.3084973 YES
## 2 4.9728091 YES 0.8097427 YES
## 3 18.5241545 NO, rerun 1.5886707 YES
## 4 6.2585948 YES 3.9769716 YES
## 5 1.8329058 YES 0.7803853 YES
## 6 0.9181118 YES 1.2657562 YES

```

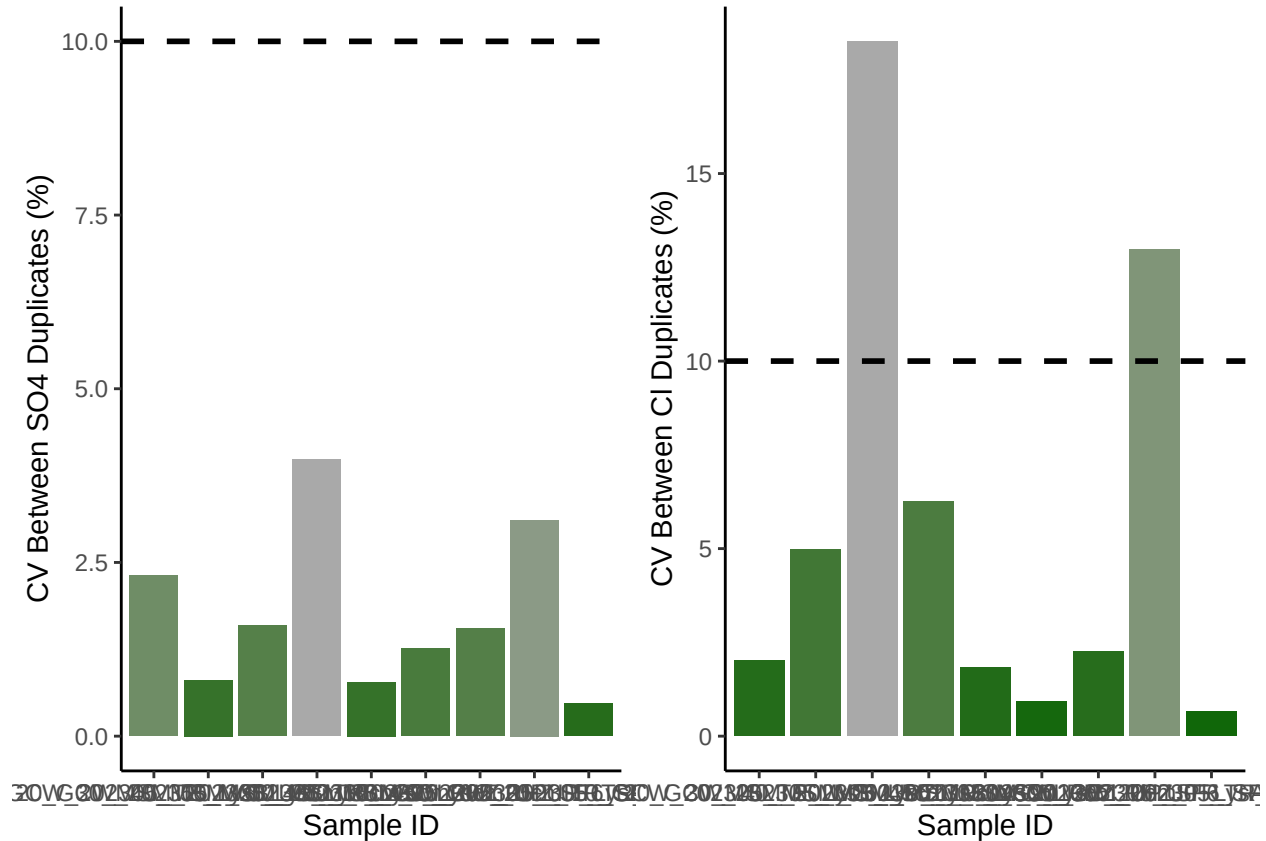
```

#plot dups output as a bar graph to easily check - want any over 10% to be red need to work on this
Sdupsbar <- ggplot(data = QAdups, aes(x = Sample_ID, y = S04_dups_cv, fill=S04_dups_chk)) +
  geom_bar(stat = 'identity') +
  scale_fill_gradient2(low='darkgreen', mid='darkgreen', high='darkgrey', space='Lab') +
  theme_classic() + labs(x= "Sample ID", y="CV Between S04 Duplicates (%)") +
  theme(legend.position="none") + geom_hline(yintercept=10, linetype="dashed",
  color = "black", size=1)

Cldupsbar <- ggplot(data = QAdups, aes(x = Sample_ID, y = Cl_dups_chk, fill=Cl_dups_chk)) +
  geom_bar(stat = 'identity') +
  scale_fill_gradient2(low='darkgreen', mid='darkgreen', high='darkgrey', space='Lab') +
  theme_classic() + labs(x= "Sample ID", y="CV Between Cl Duplicates (%)") +
  theme(legend.position="none") + geom_hline(yintercept=10, linetype="dashed",
  color = "black", size=1)

ggarrange(Sdupsbar, Cldupsbar,ncol=2, nrow=1)

```



```
#check for percent of no, reruns to see if it would warrant reruns
Perc_dups <- QAdups %>%
  group_by(SO4_dups_flag) %>%
  summarise(S_no_rows = length(SO4_dups_flag))
Perc_dups1 <- QAdups %>%
  group_by(Cl_dups_flag) %>%
  summarise(Cl_no_rows = length(Cl_dups_flag))
colnames(Perc_dups) <- c("Flag", "S_no_rows")
colnames(Perc_dups1) <- c("Flag", "Cl_no_rows")
Perc_dups2 <- cbind(Perc_dups, Perc_dups1)

Perc_dups2$Total <- length(QAdups$SO4_dups_flag)
Perc_dups2$S_Percent <- (Perc_dups2$S_no_rows / Perc_dups2$Total)*100
Perc_dups2$Cl_Percent <- (Perc_dups2$Cl_no_rows / Perc_dups2$Total)*100
head(Perc_dups2)
```

```
##   Flag S_no_rows   Flag Cl_no_rows Total S_Percent Cl_Percent
## 1  YES         9 NO, rerun      2     9      100    22.22222
## 2  YES         9   YES         7     9      100    77.77778
```

Pull out spikes and check

```
#Show me the data that we have from the calculations
head(sampledat)
```

```
##           Sample_ID S04_ppm Cl_ppm      S04_mM      Cl_mM
## 26      1_GCW_202305_UP_LysA_20cm 1.1230 0.7193 0.03502807 0.02029055
## 27      10_GCW_202305_TR_LysB_45cm 3.7603 3.1051 0.11728946 0.08759097
## 28 10_GCW_202305_TR_LysB_45cm_dup 3.8851 3.1684 0.12118216 0.08937659
## 29      11_GCW_202305_TR_LysC_10cm 2.4952 4.8296 0.07782907 0.13623695
## 30 11_GCW_202305_TR_LysC_10cm_spk 92.0921 7.9818 2.87249220 0.22515656
## 31      12_GCW_202305_TR_LysC_20cm 3.4180 6.8751 0.10661260 0.19393794
##           salinity
## 26 0.001325775
## 27 0.005636916
## 28 0.005751299
## 29 0.008753087
## 30 0.014449113
## 31 0.012449306
```

```
#pull out any rows that have "spk" in the SampleID column
spks <- sampledat %>%
  filter(str_detect(Sample_ID, "spk")) #have to change this to match data
head(spks)
```

```
##           Sample_ID S04_ppm Cl_ppm      S04_mM      Cl_mM
## 1 11_GCW_202305_TR_LysC_10cm_spk 92.0921 7.9818 2.872492 0.2251566
## 2 21_GCW_202305_WC_LysC_20cm_spk 630.4458 2680.9647 19.664560 75.6266488
## 3 31_MSM_202305_UP_LysC_10cm_spk 130.9546 462.1438 4.084672 13.0364965
## 4 41_MSM_202305_TR_LysC_20cm_spk 396.3300 3164.7510 12.362133 89.2736530
## 5 51_MSM_202305_WC_LysC_45cm_spk 225.1445 7083.8148 7.022598 199.8255233
## 6 61_MSM_202305_RHZ_UP_SF7_spk 130.4016 955.7752 4.067424 26.9612186
##           salinity
## 1 0.01444911
## 2 4.84452921
## 3 0.83511985
## 4 5.71873106
## 5 12.80047934
## 6 1.72711179
```

```
#remove the dup from these IDs so we will have duplicate sample names
spks$Sample_ID<-gsub("_spk","",as.character(spks$Sample_ID))
spks <- spks[ ,-c(2,3, 5,6)]
colnames(spks) <- c('Sample_ID', 'S04_mM_spk')
head(spks)
```

```
##           Sample_ID S04_mM_spk
## 1 11_GCW_202305_TR_LysC_10cm 2.872492
## 2 21_GCW_202305_WC_LysC_20cm 19.664560
## 3 31_MSM_202305_UP_LysC_10cm 4.084672
## 4 41_MSM_202305_TR_LysC_20cm 12.362133
## 5 51_MSM_202305_WC_LysC_45cm 7.022598
## 6 61_MSM_202305_RHZ_UP_SF7 4.067424
```

```
#put it back together with the old data set and look for duplicates
QAspks <- merge(sampdat, spks)
head(QAspks)
```

```
##           Sample_ID  S04_ppm  Cl_ppm  S04_mM  Cl_mM
## 1 11_GCW_202305_TR_LysC_10cm  2.4952  4.8296  0.07782907  0.136237
## 2 21_GCW_202305_WC_LysC_20cm 460.8081 2726.6323 14.37330318 76.914874
## 3 31_MSM_202305_UP_LysC_10cm  48.7458  504.5854  1.52045540  14.233721
## 4 41_MSM_202305_TR_LysC_20cm 303.4444 3119.0697  9.46489083  87.985041
## 5 51_MSM_202305_WC_LysC_45cm  48.6901 6942.0795  1.51871803 195.827348
## 6  61_MSM_202305_RHZ_UP_SF7  46.6181  987.7863  1.45408921  27.864212
##           salinity S04_mM_spk
## 1  0.008753087  2.872492
## 2  4.927050566 19.664560
## 3  0.911811818  4.084672
## 4  5.636184948 12.362133
## 5 12.544363656  7.022598
## 6  1.784955844  4.067424
```

```
#now we need to calculate the spike concentration and calculate the spike recovery
spkconc <- (250/smw)      # in mM
spkvol <- 10              # in uL
spkvol <- spkvol/1000000
#spike for these samples was 10uL of the 250mM standard
QAspks$S04_spk_Conc <- (spkconc)*spkvol      # mmoles of SO4
head(QAspks)
```

```
##           Sample_ID  S04_ppm  Cl_ppm  S04_mM  Cl_mM
## 1 11_GCW_202305_TR_LysC_10cm  2.4952  4.8296  0.07782907  0.136237
## 2 21_GCW_202305_WC_LysC_20cm 460.8081 2726.6323 14.37330318 76.914874
## 3 31_MSM_202305_UP_LysC_10cm  48.7458  504.5854  1.52045540  14.233721
## 4 41_MSM_202305_TR_LysC_20cm 303.4444 3119.0697  9.46489083  87.985041
## 5 51_MSM_202305_WC_LysC_45cm  48.6901 6942.0795  1.51871803 195.827348
## 6  61_MSM_202305_RHZ_UP_SF7  46.6181  987.7863  1.45408921  27.864212
##           salinity S04_mM_spk S04_spk_Conc
## 1  0.008753087  2.872492 7.797879e-05
## 2  4.927050566 19.664560 7.797879e-05
## 3  0.911811818  4.084672 7.797879e-05
## 4  5.636184948 12.362133 7.797879e-05
## 5 12.544363656  7.022598 7.797879e-05
## 6  1.784955844  4.067424 7.797879e-05
```

```
#need to determine dilution factors and initial amount of sample added
#if your samples are all the same dilution just use the first line
#for Steph / COMPASS this depends on the site so...
QAspks$Dilution <- 1
```

```
QAspks$Dilution <- ifelse(str_detect(QAspks$Sample_ID, "MSM") & str_detect(QAspks$Sample_ID, "UP"), 50,
QAspks$Dilution <- ifelse(str_detect(QAspks$Sample_ID, "MSM") & str_detect(QAspks$Sample_ID, "TR"), 50,
QAspks$Dilution <- ifelse(str_detect(QAspks$Sample_ID, "MSM") & str_detect(QAspks$Sample_ID, "WC"), 100,
QAspks$Dilution <- ifelse(str_detect(QAspks$Sample_ID, "MSM") & str_detect(QAspks$Sample_ID, "SW"), 100,
QAspks$Dilution <- ifelse(str_detect(QAspks$Sample_ID, "GCW") & str_detect(QAspks$Sample_ID, "UP"), 50,
```

```

QAspks$Dilution <- ifelse(str_detect(QAspks$Sample_ID, "GCW") & str_detect(QAspks$Sample_ID, "TR"), 50, 100)
QAspks$Dilution <- ifelse(str_detect(QAspks$Sample_ID, "GCW") & str_detect(QAspks$Sample_ID, "WC"), 100, 200)
QAspks$Dilution <- ifelse(str_detect(QAspks$Sample_ID, "GCW") & str_detect(QAspks$Sample_ID, "SW"), 100, 200)

QAspks$Dilution <- ifelse(str_detect(QAspks$Sample_ID, "GWI") & str_detect(QAspks$Sample_ID, "UP"), 100, 200)
QAspks$Dilution <- ifelse(str_detect(QAspks$Sample_ID, "GWI") & str_detect(QAspks$Sample_ID, "TR"), 100, 200)
QAspks$Dilution <- ifelse(str_detect(QAspks$Sample_ID, "GWI") & str_detect(QAspks$Sample_ID, "WC"), 200, 100)
QAspks$Dilution <- ifelse(str_detect(QAspks$Sample_ID, "GWI") & str_detect(QAspks$Sample_ID, "SW"), 200, 100)

QAspks$Dilution <- ifelse(str_detect(QAspks$Sample_ID, "SWH"), 50, QAspks$Dilution)

#Set Sample volumes in uL
QAspks$SampleVol <- 1
QAspks$SampleVol <- ifelse(QAspks$Dilution == 50, 1483, QAspks$SampleVol)
QAspks$SampleVol <- ifelse(QAspks$Dilution == 100, 1466, QAspks$SampleVol)
QAspks$SampleVol <- ifelse(QAspks$Dilution == 200, 1458, QAspks$SampleVol)

#change sample volume to L
QAspks$SampleVol <- QAspks$SampleVol/1000000
head(QAspks)

```

```

##           Sample_ID  S04_ppm  Cl_ppm      S04_mM      Cl_mM
## 1 11_GCW_202305_TR_LysC_10cm    2.4952    4.8296  0.07782907  0.136237
## 2 21_GCW_202305_WC_LysC_20cm  460.8081  2726.6323 14.37330318  76.914874
## 3 31_MSM_202305_UP_LysC_10cm   48.7458   504.5854  1.52045540  14.233721
## 4 41_MSM_202305_TR_LysC_20cm  303.4444  3119.0697  9.46489083  87.985041
## 5 51_MSM_202305_WC_LysC_45cm   48.6901  6942.0795  1.51871803 195.827348
## 6 61_MSM_202305_RHZ_UP_SF7   46.6181   987.7863  1.45408921  27.864212
##           salinity S04_mM_spk S04_spk_Conc Dilution SampleVol
## 1 0.008753087    2.872492 7.797879e-05      50 0.001483
## 2 4.927050566   19.664560 7.797879e-05     100 0.001466
## 3 0.911811818    4.084672 7.797879e-05      50 0.001483
## 4 5.636184948   12.362133 7.797879e-05      50 0.001483
## 5 12.544363656    7.022598 7.797879e-05     100 0.001466
## 6 1.784955844    4.067424 7.797879e-05      50 0.001483

```

```

#gives us the total S04 in the sample in mmoles
QAspks$S04_Total_unspkd <- (QAspks$S04_mM/QAspks$Dilution)*(QAspks$SampleVol)

##total S04 in spiked sample in mmoles
QAspks$S04_Total_spkd <- (QAspks$S04_mM_spk/QAspks$Dilution)*(QAspks$SampleVol+spkvol)

QAspks$S04_expctd_spkd <- (QAspks$S04_Total_unspkd + QAspks$S04_spk_Conc)
QAspks$spk_recovery <- (QAspks$S04_Total_spkd/QAspks$S04_expctd_spkd)*100
QAspks$S04_spks_flag <- ifelse(QAspks$spk_recovery <=120 & QAspks$spk_recovery >=80 , 'YES', 'NO', rerun=TRUE)

head(QAspks)

```

```

##           Sample_ID  S04_ppm  Cl_ppm      S04_mM      Cl_mM
## 1 11_GCW_202305_TR_LysC_10cm    2.4952    4.8296  0.07782907  0.136237
## 2 21_GCW_202305_WC_LysC_20cm  460.8081  2726.6323 14.37330318  76.914874

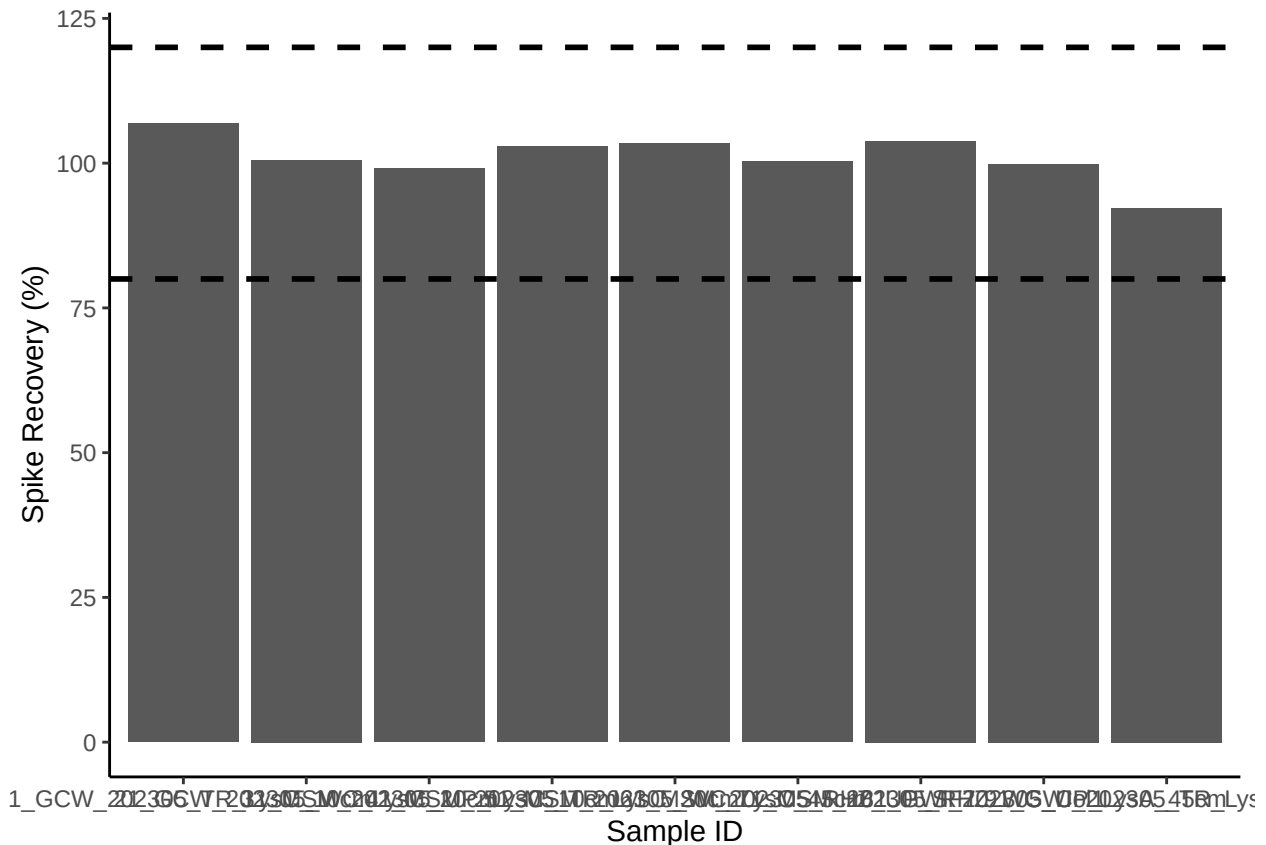
```

```
## 3 31_MSM_202305_UP_LysC_10cm 48.7458 504.5854 1.52045540 14.233721
## 4 41_MSM_202305_TR_LysC_20cm 303.4444 3119.0697 9.46489083 87.985041
## 5 51_MSM_202305_WC_LysC_45cm 48.6901 6942.0795 1.51871803 195.827348
## 6 61_MSM_202305_RHZ_UP_SF7 46.6181 987.7863 1.45408921 27.864212
##      salinity S04_mM_spk S04_spk_Conc Dilution SampleVol S04_Total_unspkd
## 1 0.008753087 2.872492 7.797879e-05 50 0.001483 2.308410e-06
## 2 4.927050566 19.664560 7.797879e-05 100 0.001466 2.107126e-04
## 3 0.911811818 4.084672 7.797879e-05 50 0.001483 4.509671e-05
## 4 5.636184948 12.362133 7.797879e-05 50 0.001483 2.807287e-04
## 5 12.544363656 7.022598 7.797879e-05 100 0.001466 2.226441e-05
## 6 1.784955844 4.067424 7.797879e-05 50 0.001483 4.312829e-05
## S04_Total_spkd S04_expctd_spkd spk_recovery S04_spks_flag
## 1 8.577262e-05 0.0000802872 106.83224 YES
## 2 2.902489e-04 0.0002886914 100.53950 YES
## 3 1.219683e-04 0.0001230755 99.10041 YES
## 4 3.691333e-04 0.0003587075 102.90651 YES
## 5 1.036536e-04 0.0001002432 103.40208 YES
## 6 1.214533e-04 0.0001211071 100.28586 YES
```

#plot spk recoveries output as a bar graph to easily check - want any over 10% to be red need to work on

```
spksbar <- ggplot(data = QAspks, aes(x = Sample_ID, y = spk_recovery)) +
  geom_bar(stat = 'identity') +
  scale_fill_manual(values=c("lightblue4")) +
  theme_classic() + labs(x= "Sample ID", y="Spike Recovery (%)") +
  theme(legend.position="none") +
  geom_hline(yintercept=80, linetype="dashed", color = "black", size=1) +
  geom_hline(yintercept=120, linetype="dashed", color = "black", size=1)
```

```
spksbar
```



```
#check for percent of no, reruns to see if it would warrant reruns
Perc_spks <- QAspks %>%
  group_by(S04_spks_flag) %>%
  summarise(no_rows = length(S04_spks_flag))
Perc_spks$Total <- length(QAspks$S04_spks_flag)
Perc_spks$Percent <- (Perc_spks$no_rows / Perc_spks$Total)*100
head(Perc_spks)
```

```
## # A tibble: 1 x 4
##   S04_spks_flag no_rows Total Percent
##   <chr>         <int> <int>   <dbl>
## 1 YES          9      9     100
```

Make final dataframe with IDs

```
#for steph <- pull out identifiers of the sample names
#pull the sample ID and separate it by the underscores
IDs <- data.frame(do.call('rbind', strsplit(as.character(sampled2$Sample_ID), '_ ', fixed=TRUE)))

## Warning in rbind(c("1", "GCW", "202305", "UP", "LysA", "20cm"), c("10", :
## number of columns of result is not a multiple of vector length (arg 16)
```



```
colnames(IDs) <- c("Analysis_No" , "Site", "Date", "Zone", "Replicate", "Depth")
head(IDs)
```

```
##   Analysis_No Site   Date Zone Replicate Depth
## 1           1  GCW 202305  UP      LysA   20cm
## 2          10  GCW 202305  TR      LysB   45cm
## 3          11  GCW 202305  TR      LysC   10cm
## 4          12  GCW 202305  TR      LysC   20cm
## 5          13  GCW 202305  TR      LysC   45cm
## 6          14  GCW 202305  WC      LysA   10cm
```

```
#rejoin them to the dataframe
alldat <- cbind(IDs, sampled2)
head(alldat)
```

```
##   Analysis_No Site   Date Zone Replicate Depth      Sample_ID
## 1           1  GCW 202305  UP      LysA   20cm 1_GCW_202305_UP_LysA_20cm
## 2          10  GCW 202305  TR      LysB   45cm 10_GCW_202305_TR_LysB_45cm
## 3          11  GCW 202305  TR      LysC   10cm 11_GCW_202305_TR_LysC_10cm
## 4          12  GCW 202305  TR      LysC   20cm 12_GCW_202305_TR_LysC_20cm
## 5          13  GCW 202305  TR      LysC   45cm 13_GCW_202305_TR_LysC_45cm
## 6          14  GCW 202305  WC      LysA   10cm 14_GCW_202305_WC_LysA_10cm
##   SO4_ppm   Cl_ppm   SO4_mM   Cl_mM   salinity
## 1   1.1230   0.7193  0.03502807  0.02029055  0.001325775
## 2   3.7603   3.1051  0.11728946  0.08759097  0.005636916
## 3   2.4952   4.8296  0.07782907  0.13623695  0.008753087
## 4   3.4180   6.8751  0.10661260  0.19393794  0.012449306
## 5  16.4178   6.8918  0.51209607  0.19440903  0.012479483
## 6 500.9219 2695.9261 15.62451341 76.04869111 4.871564463
```

Make final dataframe with IDs

```
write.csv(alldat, file="Processed Data/COMPASS_SynopticCB_PW_Processed_Cl_SO4_202305_rerun.csv") #Ch
```

END